Instructor answer key

Concepts of Biostatistics and Data Science: a two-day crash course in R

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> Instructor only. Code for every answer is in instructor/solutions.R; its full output is in instructor/outputs/solutions.log. Numbers below come from that run (R 4.6.1). Other R versions should give the same results to the decimals shown, because the data are fixed and no answer depends on random numbers except the clustering and train/test split, which use set.seed().

Running the practicals

- Before Day 1, ask participants to run R/00_install_and_check.R. On the day, check the first slide of the practical: everyone should have the project open (the RStudio title bar shows the project name).
- The most common problems: the .Rproj file was not opened, so `data/...` paths fail; lines run out of order ("object 'hearts' not found"); missing `na.rm = TRUE`; a closing bracket missing, so the console shows + and waits (press Esc).
- Beginners: keep them on the core exercises and stress interpretation. Experienced participants: point them to the extensions and ask them to explain their choice of method to a neighbour.
- Warnings such as "Removed 14 rows containing non-finite values" are expected: they are the missing CRP values. Use them to discuss reporting n.

What the simulation contains (for your judgement, not for participants)

The data were generated by data/make_hearts_data.R (seed 15). Knowing the rules helps you judge whether the methods recover them.

Feature	Rule used in the simulation
SBP at baseline	Rises 0.55 mmHg per year of age and 0.8 per kg/m2 of BMI; +5 men; +3 smokers; -1.0 per hour/week of activity; district offsets North 0, South +5, East +2, West -1; random error SD 12
Smoking	More common in men and with increasing age, less common with tertiary education
Hidden lifestyle profile	Active 30%, Average 45%, Sedentary 25% (more sedentary with age and smoking); drives activity, diet score and BMI. Stored only in instructor/hearts_truth.csv
Programme	Randomised 1:1. True effect on SBP change -5 mmHg; all participants -2 mmHg on average
CRP	Log-normal; higher with BMI and smoking
Clinic visits	Poisson rate rising with age and high BP, +0.2 log-rate for smokers, -0.15 for the programme; gamma frailty creates overdispersion
Adverse events	Risk 1% (standard) and 3.5% (programme)
Missing data	Dropout 8% completely at random; random missing values in BMI, CRP, education, activity

Two consequences are worth knowing. First, smoking also shifts people towards the sedentary profile, so its total association with SBP (about 7 mmHg after age and sex) exceeds the direct 3 mmHg built in: activity and BMI act partly as mediators of smoking. Second, the simulated programme effect is -5 mmHg; the estimates (-4.0 from change scores and from ANCOVA) are consistent with it, given their CIs.

Day 1 answers

1.1. Import the raw data

Expected output

- `dim(raw)`: 480 rows, 19 columns.

Interpretation

One row is one participant. `sbp_0` and `sbp_12` (and `high_bp_0`, `high_bp_12`) are the same measurement on the same person at baseline and 12 months, so they are paired.

Common mistakes

- Saying there are 480 "samples". There is one sample of 480 participants.
- Not noticing that two columns of the same row are paired measurements.

1.2. Missing values

Expected output

- Raw file: education 6, bmi 10, crp 14, sbp_12 41, high_bp_12 41, satisfaction 41 missing.
- `activity_min` shows a minimum of -99 in 7 rows. After `na_if(activity_min, -99)` it has 7 NAs.

Interpretation

-99 is a data-entry code for "not recorded". Left in place it would be treated as a real (impossible) value and would distort the mean and SD; with many such codes the distortion would be serious. The 41 missing 12-month values are people who left early. If leavers differ from stayers (for example, sicker people leaving), complete-case results could be biased. In this simulation dropout was random, and leavers have similar mean baseline SBP (129 mmHg in both groups), but in real data this must be checked and discussed.

Common mistakes

- Treating -99 as a real value.
- Replacing missing values with the mean.
- Not reporting how many observations an analysis used.

1.3. Variable types

Variable	Type	Variable	Type
district	Nominal	satisfaction	Ordinal (stored as numbers)
age	Continuous	clinic_visits	Count
education	Ordinal	crp	Continuous (right-skewed)
smoker	Binary	high_bp_0	Binary (derived from sbp_0)
sbp_0	Continuous	followup_months	Continuous (exposure time)

Common mistakes: calling satisfaction continuous because it is stored as numbers; calling education nominal (it is ordered). Age in whole years is still treated as continuous.

1.4. Counts and proportions

Expected output

- Smokers: 79 / 480 (16.5%).

- Tertiary education: 134 / 474 = 28.3% of those with education recorded, or 134 / 480 = 27.9% of all.

Interpretation

Either denominator can be defended. Excluding missing values describes those with known education; including them shows how much is unknown. The answer must say which is used.

Common mistakes

- Percentages without counts or denominators.
- Forgetting that `count()` shows NA as its own row.

1.5. Centre and spread

Expected output

- SBP (n = 480): mean 129.3 (SD 15.6); median 129 (IQR 118 to 140).
- CRP (n = 466): mean 3.0 (SD 3.4); median 1.9 (IQR 1.2 to 3.7).

Interpretation

SBP is roughly symmetric: mean and median agree, so mean (SD) is a good summary. CRP is right-skewed: the mean exceeds the median and the SD exceeds the mean, so report median (IQR), or analyse log(CRP).

Common mistakes

- Forgetting `na.rm = TRUE` (result NA).
- Reporting IQR as a single number without saying so; the course reports the 25th and 75th percentiles.
- Choosing the summary from a normality test.

1.6. Plots

Expected output

- BMI (n = 470): roughly symmetric, mean 26.4, median 26.2, from 16.5 to 37.6.
- CRP boxplot: smokers' box sits higher; on the log scale both groups look roughly symmetric.

Interpretation

BMI is unimodal and roughly symmetric. CRP is higher in smokers (medians 2.8 and 1.8 mg/L). The log scale spreads out the many low values and pulls in the few high ones, so the comparison is easier to see.

Common mistakes

- Using `geom_bar()` for a continuous variable.
- Leaving the default bin width without trying others.

1.7. BMI by sex (Welch t-test)

Expected output

- Women 26.3 (SD 4.0, n = 245); men 26.4 (SD 4.0, n = 225).
- Difference (women minus men) -0.10 kg/m2, 95% CI -0.83 to 0.62, p = 0.78.

Interpretation

There is no evidence of a difference in mean BMI between women and men. The CI shows the data are compatible with a difference of up to about 0.8 kg/m2 in either direction, which is small in practice. A p-value above 0.05 does not prove the means are equal; here the narrow CI is what lets us say any difference is likely to be small.

Common mistakes

- Reporting only the p-value.
- Saying "there is no difference" (absence of evidence).
- Getting the sign wrong: R reports the first factor level minus the second.

1.8. Paired change, Standard advice group

Expected output

- n = 215 with both measurements. Mean change -2.0 mmHg, 95% CI -3.2 to -0.8, p < 0.001.

Interpretation

SBP fell on average by about 2 mmHg in the standard advice group too. So the -6.0 mmHg fall in the programme group cannot all be credited to the programme: some reflects things that affect everyone (measurement, season, regression to the mean, general advice). The fair comparison is between the randomised groups: difference in mean change 4.0 mmHg (95% CI 2.3 to 5.6) in favour of the programme.

Common mistakes

- Using an independent-samples t-test on sbp_0 and sbp_12.
- Concluding the programme works from within-group change alone.

1.9. BMI by district

Expected output

- Means: North 26.0, South 27.4, East 26.5, West 25.7 kg/m2.
- Welch ANOVA: F = 4.53, p = 0.004.
- Holm-adjusted pairwise p: South vs North 0.035, South vs West 0.004; others 0.30 or more.

Interpretation

Mean BMI is not the same in all districts. The overall test does not say where the differences are. Pairwise comparisons suggest South has a higher mean than North and West, by roughly 1.4 to 1.7 kg/m2. Six comparisons are made, so p-values are adjusted to keep the overall chance of a false positive near 5%.

Common mistakes

- Running six unadjusted t-tests.
- Saying the ANOVA shows "all districts differ".

1.10. Sex and high BP

Expected output

- Women 47 / 248 (19.0%); men 77 / 232 (33.2%).
- Prevalence ratio 1.75; odds ratio 2.12 (exact 95% CI from Fisher 1.37 to 3.31).
- Chi-square (Yates) = 11.95, p < 0.001.

Interpretation

High BP was more common in men: the prevalence was about 1.75 times that in women. The OR is larger because the outcome is common. Use "prevalence" not "risk": high BP was measured at one time point (cross-sectional), not as new cases over follow-up. The chi-square test shows the association is unlikely to be due to chance alone; it does not give its size.

Common mistakes

- Using column percentages.
- Calling the OR a risk ratio.
- Inverting the comparison (women vs men).

1.11. High BP by district

Expected output

- North 20 / 119 (16.8%), South 43 / 114 (37.7%), East 38 / 125 (30.4%), West 23 / 122 (18.9%).
- Chi-square = 17.9, df = 3, $p < 0.001$.

Interpretation

Prevalence is highest in South and lowest in North. The test shows the prevalence is not the same in all districts but does not say which differ; look at the proportions, and adjust for multiple comparisons if testing pairs. Districts also differ in age and BMI, so these are crude differences.

Common mistakes

- Reading the chi-square as evidence that every district differs.

Day 1 extensions

E1.1. Kruskal-Wallis

Expected output

- Medians: 150, 145, 160 min/week; Kruskal-Wallis p = 0.55.

Interpretation

No evidence that activity tends to differ by education. The test compares whether values tend to be larger in some groups (stochastic ordering); only under equal-shaped distributions is it a comparison of medians.

Common mistakes

- Describing the test as assumption-free or as a test of medians.

E1.2. Satisfaction by sex

Expected output

- Distributions similar (e.g. 4 or 5: women 42%, men 46%); Wilcoxon p = 0.69.

Interpretation

No evidence of a difference. Satisfaction is ordinal, so means assume equal gaps between categories; show the distribution and use a rank-based method.

Common mistakes

- Reporting mean satisfaction without comment.

E1.3. Student vs Welch ANOVA

Expected output

- Classical F = 7.31, p < 0.001; Welch F = 6.83, p < 0.001.

Interpretation

Similar here because the SDs (13.4 to 17.6) and group sizes are similar. They can disagree when the group with the largest variance is much smaller or larger than the others; Welch is the safer default.

Common mistakes

- Choosing between them with a variance test p-value.

E1.4. McNemar, Standard advice group

Expected output

- Discordant cells: 17 high to not high, 12 not high to high. McNemar p = 0.46.

Interpretation

Only the discordant pairs inform change. In the standard group there is no clear net change; in the lifestyle group 28 improved and 6 worsened (p < 0.001). Comparing the groups properly would need a model (see final exercise F4).

Common mistakes

- Using chisq.test on the paired table.

E1.5. Confounding by age

Expected output

- Smokers: 8.5% of 30-49 year-olds, 23.3% of 50-74 year-olds.
- OR 2.45 (30-49) and 2.60 (50-74); Mantel-Haenszel OR 2.57 (1.51 to 4.37), versus crude 3.39.

Interpretation

Smokers are older, and older people have higher BP. Within age groups the OR is about 2.5, lower than the crude 3.4, so age confounds the crude association. Two broad age groups leave some residual confounding; the logistic model with age as continuous gives aOR 2.04.

Common mistakes

- Treating the stratum ORs as different effects (they are similar; the difference from crude is the point).

E1.6. Small counts

Expected output

- Aged 65+: 1 / 37 vs 4 / 54 events; expected counts 2.0 and 3.0; Fisher p = 0.64, OR CI 0.27 to 146.

Interpretation

Expected counts below 5 make the chi-square approximation unreliable, so Fisher's test is used. With 5 events the data are compatible with anything from a large decrease to a very large increase: the subgroup is far too small to say anything.

Common mistakes

- Interpreting the point OR (2.9) without the CI.

Day 2 answers

2.1. Scatterplot and correlation

Expected output

- Pearson r = 0.48 (95% CI 0.40 to 0.54), n = 470; Spearman rho = 0.47.

Interpretation

Moderate positive, roughly linear association: people with higher BMI tend to have higher SBP. Pearson and Spearman agree, suggesting no strong outliers or curvature. Correlation does not show causation: age and lifestyle could affect both (in this simulation BMI does raise SBP, but the data alone cannot show it).

Common mistakes

- Calling r = 0.48 "48% explained" (that would be r squared, about 23%).

2.2. Simple linear regression

Expected output

- Slope 1.87 mmHg per kg/m2 (95% CI 1.56 to 2.18); per 5 kg/m2: 9.35 (7.78 to 10.9).
- Intercept 79.9 (BMI 0, not meaningful). R-squared 0.23; residual SE 13.8 mmHg; n = 470.

Interpretation

Each 5 kg/m2 higher BMI is associated with mean SBP about 9 mmHg higher. BMI accounts for about a quarter of the variation in SBP. Residual plots show no curve or funnel and the Q-Q plot is close to the line: the assumptions look reasonable. Independence comes from the design (one row per person).

Common mistakes

- Interpreting the intercept.
- Checking normality of SBP instead of the residuals.
- Saying R-squared is low so the association is unimportant.

2.3. Confounding

Expected output

- Crude: smokers 12.5 mmHg higher (8.9 to 16.1).
- Adjusted for age and sex: 7.0 mmHg (3.7 to 10.3).

Interpretation

Smokers are older and more often male, and both raise SBP, so part of the crude difference was due to age and sex. "Among people of the same age and sex, smokers had a mean SBP about 7 mmHg higher than non-smokers (95% CI 3.7 to 10.3)." CRP is raised by smoking; it is not a cause of smoking, so it is not a confounder. If CRP lay on the path from smoking to SBP it would be a mediator, and adjusting would remove part of the effect of interest; if it does not affect SBP, adjusting adds nothing useful. Either way, do not adjust for it when estimating the total association.

Common mistakes

- Choosing adjustment variables by p-value.
- Adjusting for everything available.
- Interpreting the adjusted estimate as causal without discussing residual confounding.

2.4. Plot a fitted relationship

Expected output

- Model sbp_0 ~ bmi + age + sex: BMI 1.42 (1.14 to 1.71) mmHg per kg/m2; age 0.52 per year; men +5.4.

Interpretation

Two parallel lines rising with BMI, men about 5 mmHg higher at every BMI. The model assumes the same BMI slope in women and men (no interaction). The BMI slope is smaller than in 2.2 (1.87) because older people have slightly higher BMI and higher SBP.

Common mistakes

- Plotting predictions outside the observed BMI range.
- Forgetting to fix the other variables.

2.5. Logistic regression

Expected output

- aOR per 5 kg/m2 BMI 2.92 (95% CI 2.08 to 4.18); per 10 years age 2.13 (1.70 to 2.70); men 2.52 (1.54 to 4.19).

Interpretation

Among people of the same age and sex, each 5 kg/m2 higher BMI was associated with about 2.9 times the odds of high BP. Because high BP is common (26%), the ratio of probabilities is smaller than the OR, so "times the risk" would overstate it. Predicted probabilities are a clearer way to communicate it.

Common mistakes

- Reporting raw coefficients (log-odds) as ORs.
- Forgetting `exponentiate = TRUE`.
- Calling the aOR a risk ratio.

2.6. Poisson regression

Expected output

- Poisson: IRR smokers 1.24 (1.08 to 1.41); age per 10 years 1.23.
- Dispersion ratio 2.06. Quasi-Poisson: IRR 1.24 (1.02 to 1.50).

Interpretation

Adjusting for age, smokers had about 24% more unplanned visits per month of follow-up. The offset is needed because follow-up ranged from 2 to 12 months: raw counts mix the rate with the time at risk. The dispersion ratio of about 2 means the counts vary twice as much as Poisson assumes, so Poisson CIs are too narrow; quasi-Poisson keeps the IRR and widens the CI.

Common mistakes

- Omitting the offset, or entering follow-up as an ordinary covariate without the log.
- Ignoring overdispersion.

2.7. k-means with k = 2 and k = 3

Expected output

- k = 2: cluster of 215 (median activity 255 min/week, diet 6.8, BMI 24.1) and 248 (85 min/week, diet 4.4, BMI 28.4).
- k = 3: 146 (60 min/week, diet 3.9, BMI 30.2), 228 (165, 5.8, 25.1), 89 (390, 7.3, 23.5).
- Silhouette widths: 0.34 (k = 2), 0.29 (k = 3).

Interpretation

Names such as "less active, higher BMI" and "more active, healthier diet" (k = 2), with a middle group added at k = 3. Variables were scaled because activity is in hundreds of minutes and would otherwise dominate the distances; unscaled clusters were just bands of activity. Reasons for caution: low silhouette (weak structure); results change with k, variables, scaling and algorithm; clustering always returns clusters, even from data without groups; no external validation yet.

Common mistakes

- Treating cluster numbers as meaningful labels.
- Testing whether clusters differ on the variables used to build them.
- Forgetting set.seed() and nstart.

Instructor note: compared with the hidden simulated profile, the k = 3 cluster of 89 is almost entirely Active (85), the cluster of 146 is mostly Sedentary (102 of 146), and the middle cluster of 228 contains 160 Average, 55 Active and 13 Sedentary. The clusters capture the main gradient but do not reproduce the true groups exactly: a good illustration of the caution on the slides.

Day 2 extensions

E2.1. Change scores vs ANCOVA

Expected output

- Change scores: difference 4.0 mmHg (95% CI 2.3 to 5.6). ANCOVA: -4.0 (-5.6 to -2.3).

Interpretation

Almost identical here because the baseline coefficient is close to 1 (1.01). In general ANCOVA is at least as precise and is the usual primary analysis for a randomised trial with a baseline measurement. The sign differs only because of how the comparison is set up.

Common mistakes

- Comparing follow-up values without baseline when baseline is available.

E2.2. Train and test

Expected output

- sbp_0 ~ age + sex + bmi: RMSE 11.9 (training), 12.5 (test). Age only: 13.4 / 13.7. Six predictors: 11.4 / 12.3.

Interpretation

Training error is optimistic because the model was fitted to those data. Test error estimates performance on new people. Leakage examples: selecting predictors using all data before splitting; scaling with the mean and SD of the whole dataset; predicting high_bp_12 using sbp_12.

Common mistakes

- Evaluating on the training data.

E2.3. Negative binomial

Expected output

- Programme IRR 0.93 (0.79 to 1.08); Poisson 0.91 (0.82 to 1.02); quasi-Poisson 0.91 (0.78 to 1.06).

Interpretation

All give similar IRRs; the overdispersion-aware models give wider CIs. None shows clear evidence that visit rates differ between programme groups.

Common mistakes

- Choosing the model that gives the smallest p-value.

E2.4. Linkage methods

Expected output

- Average linkage, 3 groups: 426, 30, 7. Ward: 121, 152, 190.

Interpretation

Average linkage separates a few outlying people and leaves almost everyone in one group; Ward's method favours compact groups of similar size. The "clusters" depend heavily on the method.

Common mistakes

- Assuming hierarchical clustering gives one objective answer.

Final exercise: model answers and marking guide

Look for: a stated question; correct outcome type and design; a justified method; a suitable plot; an estimate with CI; a check of assumptions; a plain-language conclusion with a limitation. Accept any defensible method that is justified.

Element	Strong answer	Weak answer
Method choice	Linked to question, outcome type and design	"Data not normal so nonparametric"
Result	Estimate, 95% CI, n	p-value only
Conclusion	Size, uncertainty, practical meaning, limitation	"Significant" / "not significant" only
Causation	Causal language only for the randomised comparison	Causal claims from observational data

F1. Satisfaction by programme

Expected output

- Satisfied (4 or 5): standard 71 / 215 (33.0%); programme 123 / 224 (54.9%). Very dissatisfied (1): 11.2% vs 4.9%.
- Wilcoxon rank-sum $p < 0.001$ ($W = 17969$). Plot: stacked or side-by-side bar chart of percentages at each level.

Interpretation

Participants randomised to the programme tended to report higher satisfaction: 55% were satisfied or very satisfied compared with 33% with standard advice. Because allocation was random, the difference can reasonably be attributed to the programme, though participants were not blinded, which may affect a self-reported outcome. 41 people without follow-up are excluded.

Common mistakes

- Reporting mean satisfaction as the main result.
- Ignoring the 41 missing responses.

F2. CRP by sex

Expected output

- Medians: women 1.85 (IQR 1.10 to 3.62, n = 240), men 2.00 (1.20 to 3.77, n = 226).
- Wilcoxon p = 0.50; Hodges-Lehmann shift -0.10 mg/L (-0.40 to 0.20).
- Log scale: geometric mean ratio men/women 1.08 (95% CI 0.92 to 1.27).

Interpretation

There is no clear difference in CRP between men and women. On the log scale, men's typical CRP is estimated 8% higher, but the data are compatible with anything from 8% lower to 27% higher. Plot: boxplots on a log scale with points.

Common mistakes

- Using a t-test on raw CRP without comment.
- Describing the Wilcoxon result as a difference in medians.

F3. Diet score and SBP

Expected output

- Crude: -2.32 mmHg per diet point (-3.08 to -1.55).
- Adjusted for age and sex: -1.54 (-2.21 to -0.87).

Interpretation

Among people of the same age and sex, each extra point on the diet score was associated with mean SBP about 1.5 mmHg lower (a 4-point difference, roughly the IQR, corresponds to about 6 mmHg). Part of the crude association was due to age and sex. This is observational: diet may stand for other aspects of lifestyle (activity, BMI), some of which may be mediators. Plot: scatterplot with jitter and fitted line.

Common mistakes

- Adjusting for BMI without discussing that it may be a mediator.
- Causal wording.

F4. High BP at 12 months by programme

Expected output

- Standard 50 / 215 (23.3%); programme 35 / 224 (15.6%); risk difference -7.6 percentage points; risk ratio 0.67 (95% CI 0.46 to 0.99, from the log-RR formula).
- Chi-square with Yates correction p = 0.057.
- Logistic model adjusting for baseline high BP: OR 0.46 (0.25 to 0.85).

Interpretation

At 12 months, high BP was less common in the programme group (16% vs 23%). The unadjusted test is borderline, but adjusting for baseline status (which strongly predicts follow-up status) gives a more precise estimate that clearly favours the programme. Because allocation was random this supports a benefit of the programme, but 41 participants with no follow-up are excluded, and dichotomising SBP loses information: the continuous analysis (E2.1) is more informative.

Common mistakes

- Treating p = 0.057 as proof of no effect.
- Using McNemar's test to compare the two groups (McNemar is for paired data within one group).
- Forgetting the missing follow-up.